FreeFlow-ONT User Manual

(version 1.0)

- FreeFlow-ONT is a Galaxy-based framework for the analysis of Oxford Nanopore Technologies direct RNA sequencing (ONT DRS) data. It provides an integrated and user-friendly environment that supports key steps of ONT DRS analysis, including data preprocessing, alignment, transcript-level analysis, and downstream functional exploration. The framework is designed to improve accessibility, standardization, and efficiency in ONT DRS data analysis for both routine and advanced research applications.
- FreeFlow-ONT was powered with an advanced packaging technology, which enables compatibility and portability.
- FreeFlow-ONT project is hosted on https://github.com/jy-ai/FreeFlow-ONT
- FreeFlow-ONT docker image is available at https://hub.docker.com/r/malab/freeflowont

Functional Annotation Module

This module is designed to provide functional interpretation for assembled transcripts. It includes two steps: **coding potential and ORF annotation** and **functional annotation**. In the first step, **TranslationAI** is used to predict translation initiation and termination related ORF information from transcript sequences. In the second step, predicted transcript-derived protein sequences are functionally annotated using **eggNOG-mapper** based on homology.

Tools	Description	Input	Output	Time (test data)	Reference
Step1: Coding Potential & ORF Annotation	Perform TranslationAI-based ORF annotation for assembled transcripts	Assembled transcript FASTA	TranslationAI prediction results	Depends on the dataset size	TranslationAI
Step2: Functional Annotation	Perform homology-based functional annotation for assembled transcripts	Assembled transcript FASTA, transcript annotation TXT	Eggnog annotation results	Depends on the dataset size	eggNOG-mapper

Step1: Coding Potential & ORF Annotation

This function is designed to predict coding potential and ORF-related features from assembled transcript sequences using **TranslationAI**. It analyzes transcript sequences to identify translation initiation sites and termination sites, and generates annotation results in TXT format.

Input

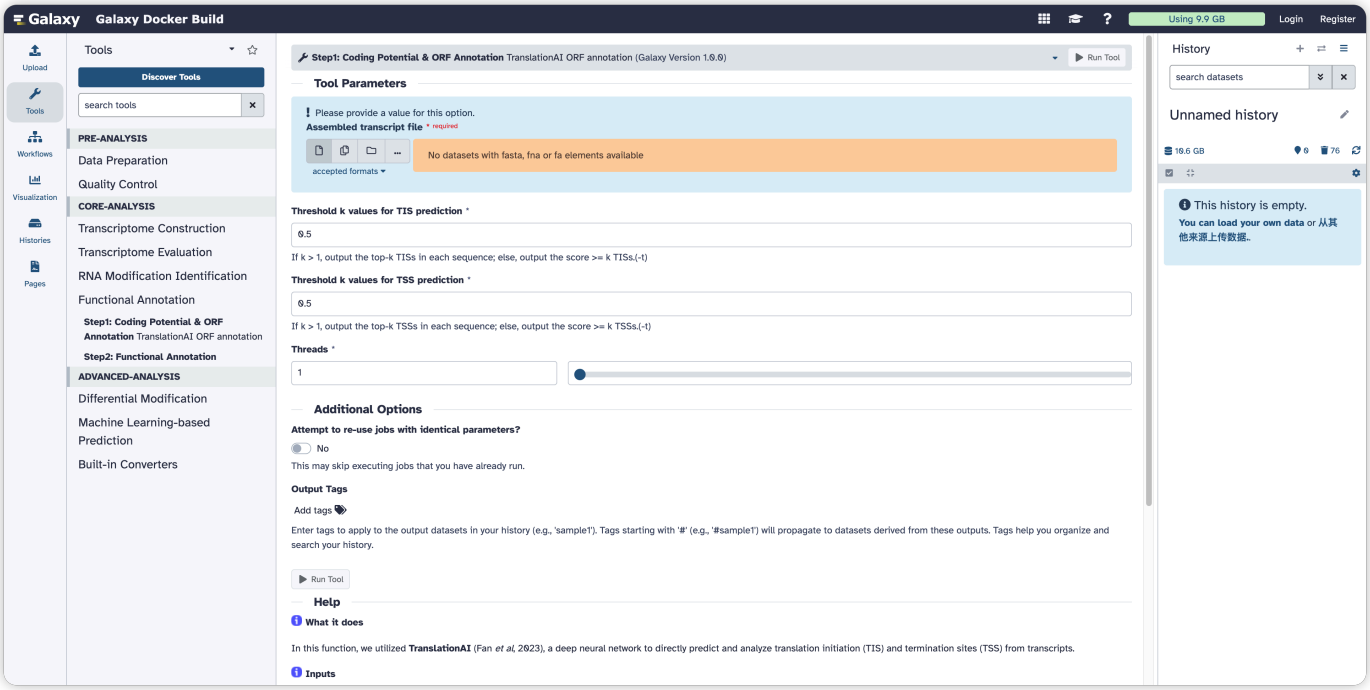
- **Assembled transcript file:** A FASTA file containing assembled transcript sequences.

Parameters

- **Threshold k values for TIS prediction:** A float specifying the threshold for translation initiation site prediction. If the value is greater than 1, the top-k TISs in each sequence are reported; otherwise, sites with scores greater than or equal to the threshold are reported. The default value is **0.5**.
- **Threshold k values for TSS prediction:** A float specifying the threshold for translation termination site prediction. If the value is greater than 1, the top-k TSSs in each sequence are reported; otherwise, sites with scores greater than or equal to the threshold are reported. The default value is **0.5**.
- **Threads:** An integer specifying the number of CPU threads available for the analysis. The default value is **1**.

Output

- **TranslationAI Prediction Results:** A TXT file containing predicted ORF-related annotation results.



Step2: Functional Annotation

This function is designed to perform functional annotation for assembled transcripts using **eggNOG-mapper**. It takes assembled transcript sequences together with a transcript annotation file as input, translates transcripts into peptide sequences through the internal workflow, and identifies potential functions based on homology.

Input

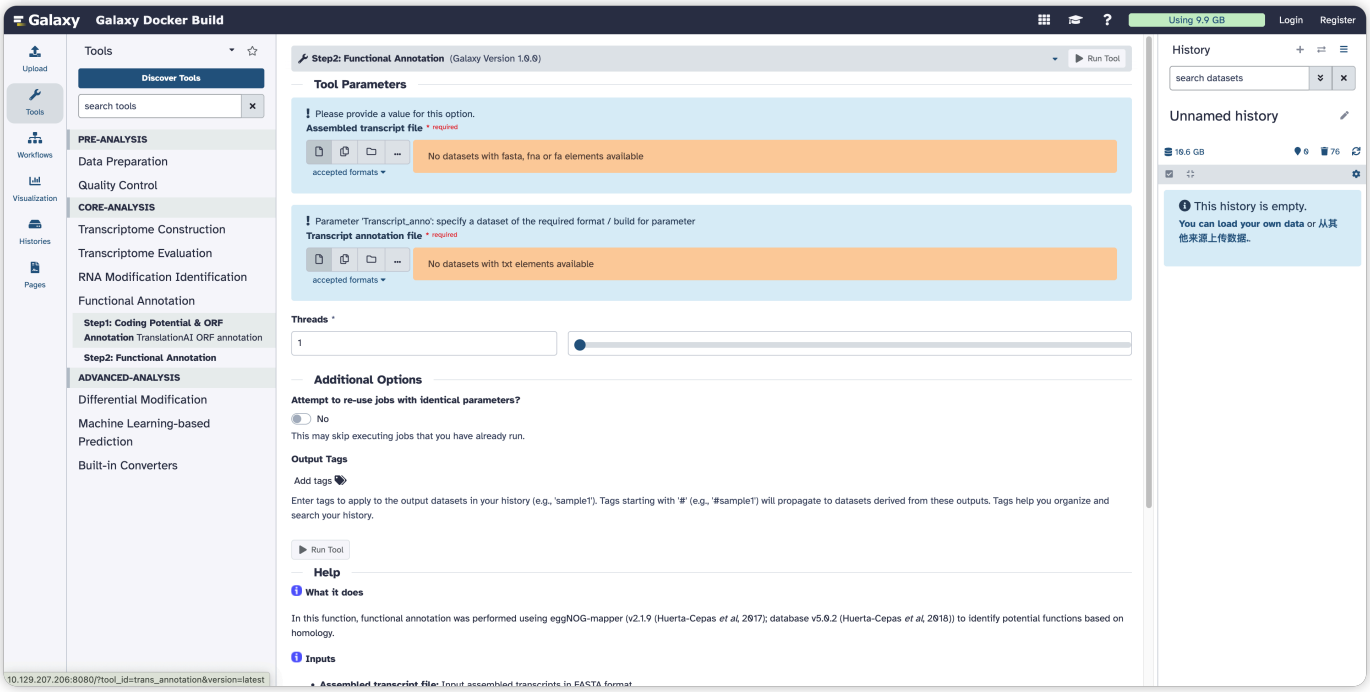
- **Assembled transcript file:** A FASTA file containing assembled transcript sequences.
- **Transcript annotation file:** A TXT file containing transcript annotation results, which can be used as input for downstream functional annotation.

Parameters

- **Threads:** An integer specifying the number of CPU threads used during eggNOG-mapper annotation. The default value is **1**.

Output

- **Eggnog Annotation Results:** A TXT file containing functional annotation results generated by eggNOG-mapper.



Notes

- The first step focuses on **coding potential and ORF-related prediction**, while the second step provides **homology-based functional annotation** for the resulting transcript-derived protein sequences.
- The **Transcript annotation file** used in Step2 is intended to work with the assembled transcript FASTA and can serve as the annotation input for the functional annotation workflow.